ANUBHAV SEN

linkedin.com/in/anubhav-sen-6b7939239 | (484) 705-9666 | asen4.github.io | anubhav.4.sen@gmail.com | github.com/asen4

EDUCATION

Bachelor of Science in Computer Engineering

The Pennsylvania State University (University Park, PA) — May 2026

College of Engineering | Schreyer Honors College — Cumulative GPA: **3.94/4.0**

TECHNICAL SKILLS

- Proficient: C, Java, Python, Git, Android SDK, Android Studio, Dash, Google Firebase, XML, Verilog
- Intermediate: SQL, JavaScript, HTML, CSS, Node.js, MongoDB, React Native, Shiny

WORK EXPERIENCE

Cybersecurity Intern (SEPTA) — **June 2026 – August 2026**

- Identified **2,000+ instances of employee data** exposed across public sources by running open-source intelligence (OSINT) scans with theHarvester and SpiderFoot, mapping our organization's external attack surface to reduce the risk of future phishing attacks.
- Applied spaCy for named entity recognition and Levenshtein distance matching to automatically clean the scraped employee data, identifying and merging roughly **30% of entries** as duplicates from overlapping sources.
- Grouped the records into **18 clusters** using TF-IDF vectorization and K-means, organizing findings by department and job function to help determine which teams were the most vulnerable and had the highest concentration of publicly exposed data.

Global Data Operations Intern (Merck Sharp & Dohme) — **June 2024 – August 2024**

- Developed a streamlined, user-friendly dashboard to centralize data automation tools, significantly improving accessibility and usability while **reducing the time spent running manual tasks by 30%** for **over 500 users globally**.
- Created a one-click interface for running these tools, garnering strong support from non-technical users and eliminating the need for any prior experience using the command shell or knowing system-level commands.
- Incorporated the Dash library and collaborated with our IT department to secure access for deploying the site on AWS.

SOFTWARE DEVELOPMENT PROJECTS

React Native Development — **May 2023 – Present**

- ***Nittany Retails***: e-commerce (JavaScript-based) application — **August 2023**
  - Architected and hosted Node.js/Express backend, powered by MongoDB to manage listings, user accounts, and transactions.
  - Integrated the Stripe API to process and record user purchases with payment intents, keeping card details off our server.

Android Application Development — **August 2019 – Present**

- Published **3 Android (Java-based) mobile applications** to the Google Play Store that have a growing user base while currently exploring new ways to boost engagement and iteratively improve features based on user feedback.
- ***Smart Planner***: digital student planner application — **July 2022**
  - Implemented a student planner that syncs Canvas assignments to Google Calendar (via its API) and sends automated reminders, so students can track deadlines across devices.
- ***TalkZone***: video-calling/casting media application — **October 2021**
  - Integrated the open-source Jitsi video conferencing SDK with the Google Cast API to support real-time video calls and casting from mobile devices to TV screens.
- ***TagOut!***: social media application — **July 2021**
  - Developed key features such as phone-based authentication, live messaging, and personalized user feeds, powered by Google Firebase's real-time database and cloud functions for reliable performance at scale.

RESEARCH EXPERIENCE

Chimeric and Outward Reads: A Unified Circular RNA Assembly Framework — **February 2025 – November 2026**

- Outperformed 4 circular RNA assemblers on F-score, with **28% higher recall** than the runner-up **at comparable precision (~41%)**, by building a graph of candidates from sequencing reads and screening them with 2 random forests and a gradient-boosted model.
- Improved **F-score 3.5%** on noisier input while prior method fell 19%, by greedily selecting candidates that best explain the reads.

AI for Politics — **November 2022 – November 2024**

- Predicted future US election outcomes by scraping 10K tweets/day (X API) and training a logistic-regression text classifier using TF-IDF features to label political leaning, then aggregated results into a color-coded heat map in Folium to showcase regional trends.

LEADERSHIP/ACTIVITIES

AlgoPSU Director — **January 2023 – May 2026**

- Mentored **25 students** on system design principles and walked through LeetCode-style questions to build interview readiness.

Java: A Comprehensive Guide (Parts I & II) — **October 2021 – August 2022**

- Authored and published a two-part written guide giving a primer on fundamental data structures, algorithms, and core programming concepts in Java; only available on Amazon as a Kindle eBook, hardcover, and paperback.